

Summary – Customer Churn Analysis

This analysis focuses on understanding customer churn patterns and identifying the primary drivers influencing customer retention. The dataset was cleaned and prepared by handling missing values in *Total Charges* (mainly for customers with zero tenure) and converting categorical variables into interpretable formats. Exploratory Data Analysis (EDA) was conducted using well-structured visualizations to uncover meaningful trends.

Overall Churn Overview

- The overall **customer churn rate is approximately 26.5%**, meaning **nearly 1 out of every 4 customers leaves the company**.
 - This indicates a **moderate to high churn risk**, emphasizing the need for focused retention strategies.
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Demographic Insights

- **Senior citizens show a significantly higher churn rate (~42%)** compared to **non-senior customers (~24%)**.
 - This gap suggests that older customers may face usability, pricing, or service-related challenges that are not being adequately addressed.
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Tenure-Based Behavior

- Customers with **short tenure (0–12 months)** exhibit the **highest churn rate, close to 45–50%**.
 - In contrast, customers who stay beyond **24 months show churn rates below 15%**.
 - This clearly indicates that **early-stage customers are the most vulnerable**, making onboarding and first-year engagement critical.
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Contract Type Impact

Contract duration is one of the **strongest predictors of churn**:

- **Month-to-month contracts** show the **highest churn rate (~43%)**
- **One-year contracts** have a significantly lower churn rate (**~11%**)
- **Two-year contracts** demonstrate the strongest retention, with churn as low as **~3%**

Long-term contracts clearly enhance customer loyalty and stability.

Payment Method Analysis

- Customers using **Electronic Check** experience the **highest churn (~45%)**
 - Other payment methods such as **credit cards, bank transfers, and mailed checks** have churn rates **below 20%**
 - This suggests that electronic check users may face **payment friction, trust issues, or inconsistent billing experiences**.
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Service & Product Insights

- Customers using **Fiber Optic internet** show a **higher churn rate (~40%+)** compared to DSL users.
- However, churn drops significantly when customers subscribe to **value-added services**:
 - **Technical Support**
 - **Online Security**
 - **Online Backup**

Customers **without these support services** churn nearly twice as much as those who have them, indicating that **bundled services increase perceived value and retention**.

Key Business Takeaways

- Churn is primarily driven by:
 - Short customer tenure
 - Month-to-month contracts
 - Electronic check payment method
 - Lack of support and security services
 - **Early engagement (first 6–12 months)** is the most critical intervention window.
 - Encouraging **long-term contracts** and **service bundling** can significantly reduce churn.
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Conclusion

To reduce churn, the company should focus on:

- Strengthening **customer onboarding and first-year experience**
- Incentivizing customers to switch from **month-to-month to long-term contracts**
- Improving or replacing **electronic check payment experiences**
- Promoting **technical support and security add-ons**, especially for fiber optic users